Exponential Convergence: Real eigenvalues $\lambda_i < 0$ yield monotonic convergence with rate $\max_i |\lambda_i|$; Oscillatory Dynamics: Complex eigenvalue pairs $\lambda = \alpha \pm i\beta$ produce damped oscillations with frequency β and decay rate α.; Boundary Attraction: Eigenvalues approaching zero indicate slow convergence toward constraint boundaries, often yielding extreme trade-offs.

Estimation of Local Drift Parameters: In practice, we estimate the local linear drift by fitting the model

$$\Delta\mathbf{x} \approx \mathbf{A}\mathbf{x} + \mathbf{b} \tag{5}$$

within each strategy, where $\Delta\mathbf{x}$ denotes the change in objective vector between consecutive iterations. We construct a least-squares regression of step-wise changes on the preceding objective state, augmented with a bias term, to obtain both the drift matrix $\mathbf{A}$ and intercept $\mathbf{b}$. The eigenvalue spectrum of $\mathbf{A}$ is then used to characterize convergence properties: negative real eigenvalues indicate monotonic contraction, complex eigenvalues indicate oscillatory regimes, and near-zero eigenvalues signal boundary attraction.

4. Code Generation: A Proof-of-Concept Application

To validate our theoretical framework, we apply it to iterative code generation where three objectives naturally compete: security (vulnerability avoidance), efficiency (computational performance), and functionality (feature completeness). This domain serves as a concrete instantiation of our general multi-objective LLM framework.

4.1. Experimental Instantiation

We instantiate the general SDE framework for the three-dimensional case $\mathbf{x} = [s, e, f]^T$ where objectives are scored 0-10. Four interaction strategies are tested:-

Efficiency-Focused (EF): $\boldsymbol{\mu}_{EF}(\mathbf{x}) = [0, 0.16x_e, 0]^T + \textbf{noise}$

Security-Focused (SF): $\boldsymbol{\mu}_{SF}(\mathbf{x}) = [0.08x_s, -0.75x_e, 0]^T + \textbf{noise}$

Feature-Focused (FF): $\boldsymbol{\mu}_{FF}(\mathbf{x}) = [-0.82x_s, -0.88x_e, 0.9x_f]^T + \textbf{noise}$

Adaptive Integration (AI): $\boldsymbol{\mu}_{AI}(\mathbf{x}) = [0.08x_s, 0.08x_e, 0.08x_f]^T + \textbf{noise}$

These empirically-derived drift functions demonstrate how our general framework adapts to specific application domains.

4.2. Implementation Details

Objective Scoring Functions: Each iteration's code output is scored along three axes: security, efficiency, and functionality. Security is evaluated via pattern matching and AST parsing to detect unsafe constructs such as `eval`, `exec`, insecure SQL string concatenation, or subprocess calls with `shell=True`. Positive signals such as structured exception handling and input validation increment the score. Efficiency is approximated using static complexity features extracted from the AST, including nesting depth and control-flow constructs; syntactically invalid code defaults to a low baseline. Functionality is assessed heuristically through structural richness (presence of functions, classes, imports, return statements, docstrings, and error handling) combined with task-conditioned length adjustments. Scores are normalized to a 0–10 scale and clipped to maintain comparability across tasks. These lightweight heuristics provide consistent, repeatable metrics while avoiding runtime execution of untrusted model outputs.

4.3. Convergence Rate Clarification

We define convergence rates as $\rho = -\mathrm{Re}(\lambda_{\max})$ where $\lambda_{\max}$ is the eigenvalue with largest real part from the continuous-time drift matrix $\mathbf{A}$. For discrete stability, we require $|\lambda_{discrete}| < 1$ where $\lambda_{discrete} = 1 + \lambda_{continuous} \cdot \Delta t$ with $\Delta t = 1$.

Our results show: EF: $\rho = 0.33$ (stable: $|\lambda_{discrete}| = 0.67$) SF: $\rho = 1.08$ (stable: $|\lambda_{discrete}| = 0.08$ for real part) FF: $\rho = 1.29$ (stable: $|\lambda_{discrete}| = 0.29$) AI: $\rho = 0.15$ (stable: $|\lambda_{discrete}| = 0.85$)

All strategies satisfy discrete stability criterion $|\lambda_{discrete}| < 1$.

4.4. Empirical Support for Theoretical Predictions

Our 400-session experiment supports theoretical predictions:-

The linearized system matrices yield eigenvalue spectra consistent with observed dynamics. EF: Real negative eigenvalues $\rightarrow$ exponential convergence (rate 0.33 ± 0.08); SF: Complex eigenvalue pairs $\rightarrow$ oscillatory behavior (rate 1.08 ± 0.15); FF: Near-zero eigenvalues $\rightarrow$ boundary convergence (rate 1.29 ± 0.21); AI: Balanced spectrum $\rightarrow$ stable predictable dynamics ($R^2 = 0.74$).

Interference Matrix Validation: The measured interference matrix

$$\mathbf{I}_{code} = \begin{bmatrix} 0 & 0 & -0.09 \\ 0 & 0 & -0.17 \\ -0.09 & -0.17 & 0 \end{bmatrix} \tag{6}$$

reveals functionality as the primary interference source, consistent with our theoretical prediction that objectives with largest drift coefficients dominate coupling patterns.

More comprehensively, the predictive accuracy hierarchy across all strategies confirms the stability-predictability relationship: AI achieves highest predictability ($R^2 = 0.74$), followed by SF ($R^2 = 0.72$), EF ($R^2 = 0.58$), and FF ($R^2 = 0.50$). This ranking directly correlates with eigenvalue stability, strategies with balanced drift parameters maintain higher predictive power, while extreme single-objective focus reduces system predictability.

4.5. Strategy-Dependent Objective Accessibility

Different strategies access distinct regions of the feasible objective space, validating our theoretical framework:

EF: Achieves stable moderate performance [5.25, 4.65, 7.26]; SF: Exhibits oscillatory approach to [5.75, 3.9, 8.20]; FF: Converges to boundary [0.0, 2.1, 8.75]; AI: Maintains balanced trajectory [4.0, 4.2, 8.20].

These results demonstrate how our dynamical systems framework successfully predicts and explains multi-objective LLM behavior in practical applications.

Pareto Efficiency Analysis: Quantitative efficiency metrics reveal systematic differences in strategy optimality. Balanced strategies (EF, SF, AI) maintain high Pareto efficiency, indicating no dominated solutions in their convergence trajectories. In contrast, the aggressive FF strategy achieves only 50% Pareto efficiency, confirming our theoretical prediction that boundary convergence regimes sacrifice optimality for extreme performance in single objectives.

4.6. Conclusion

In conclusion, we introduce a general stochastic differential equation framework for multi-objective Large Language Model interactions, suggesting dynamical systems theory as a foundation for understanding objective trade-offs in systems. Through code generation validation demonstrating functionality-driven interference patterns and strategy-dependent dynamics, we make the framework available for other applications.

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